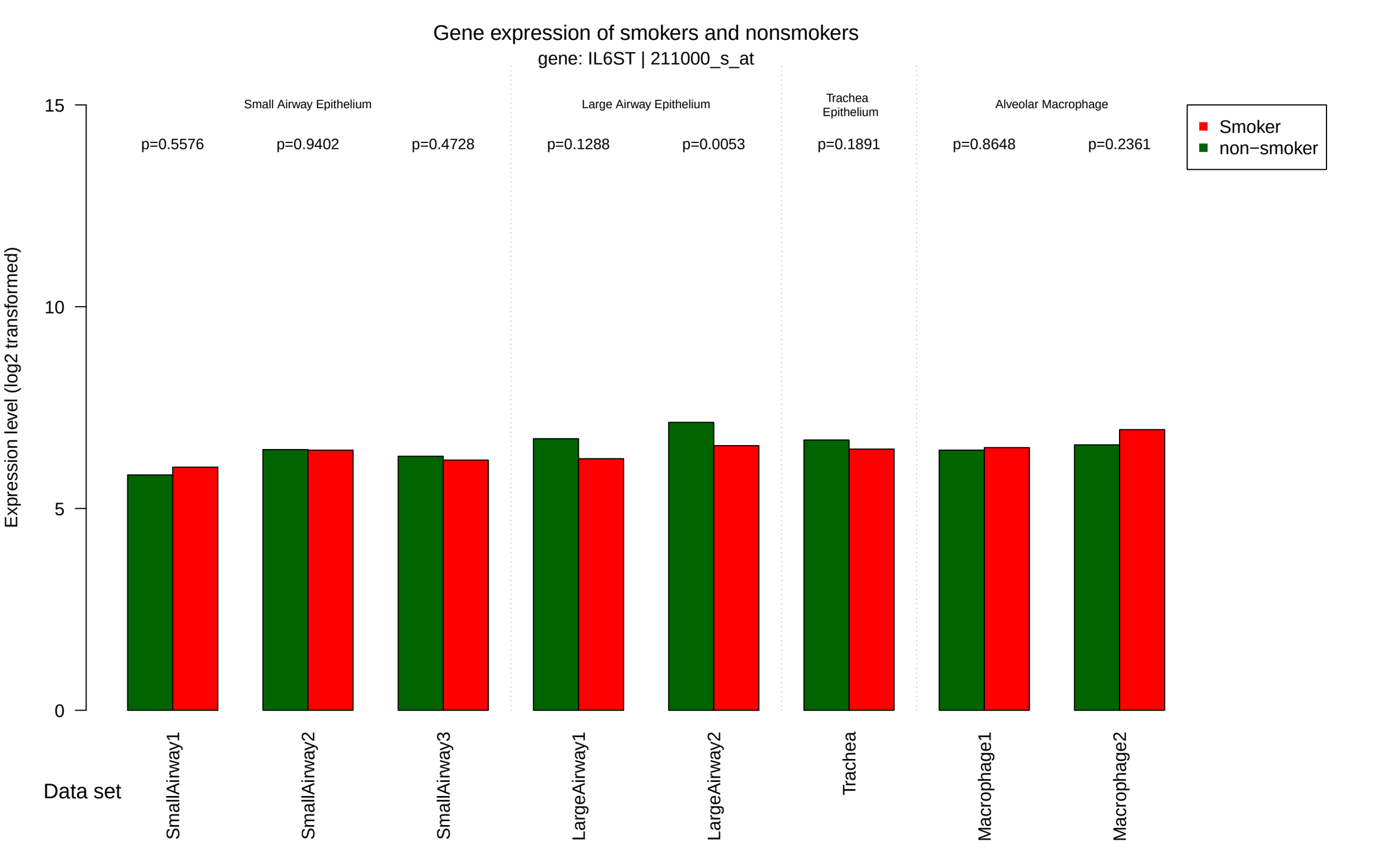
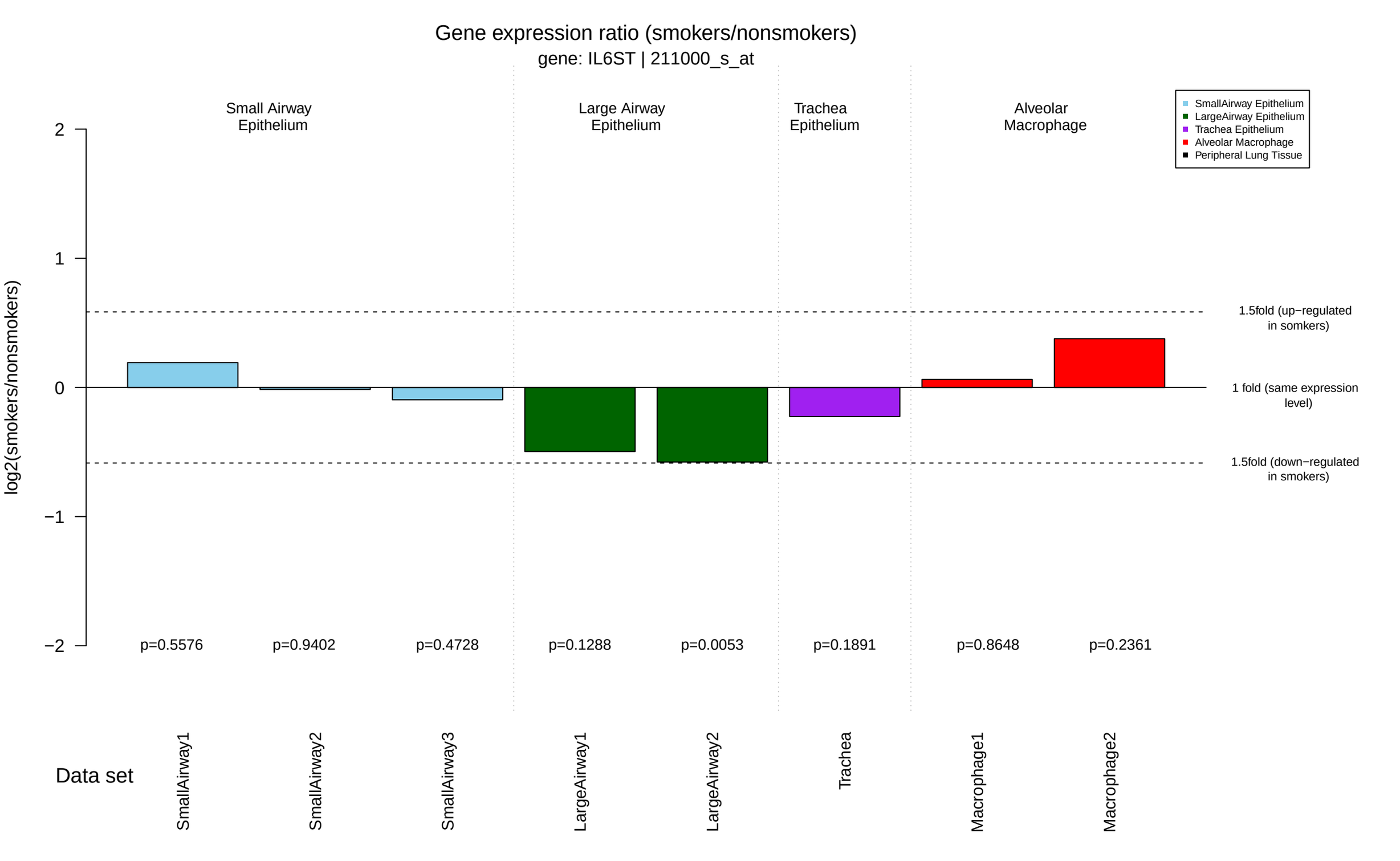
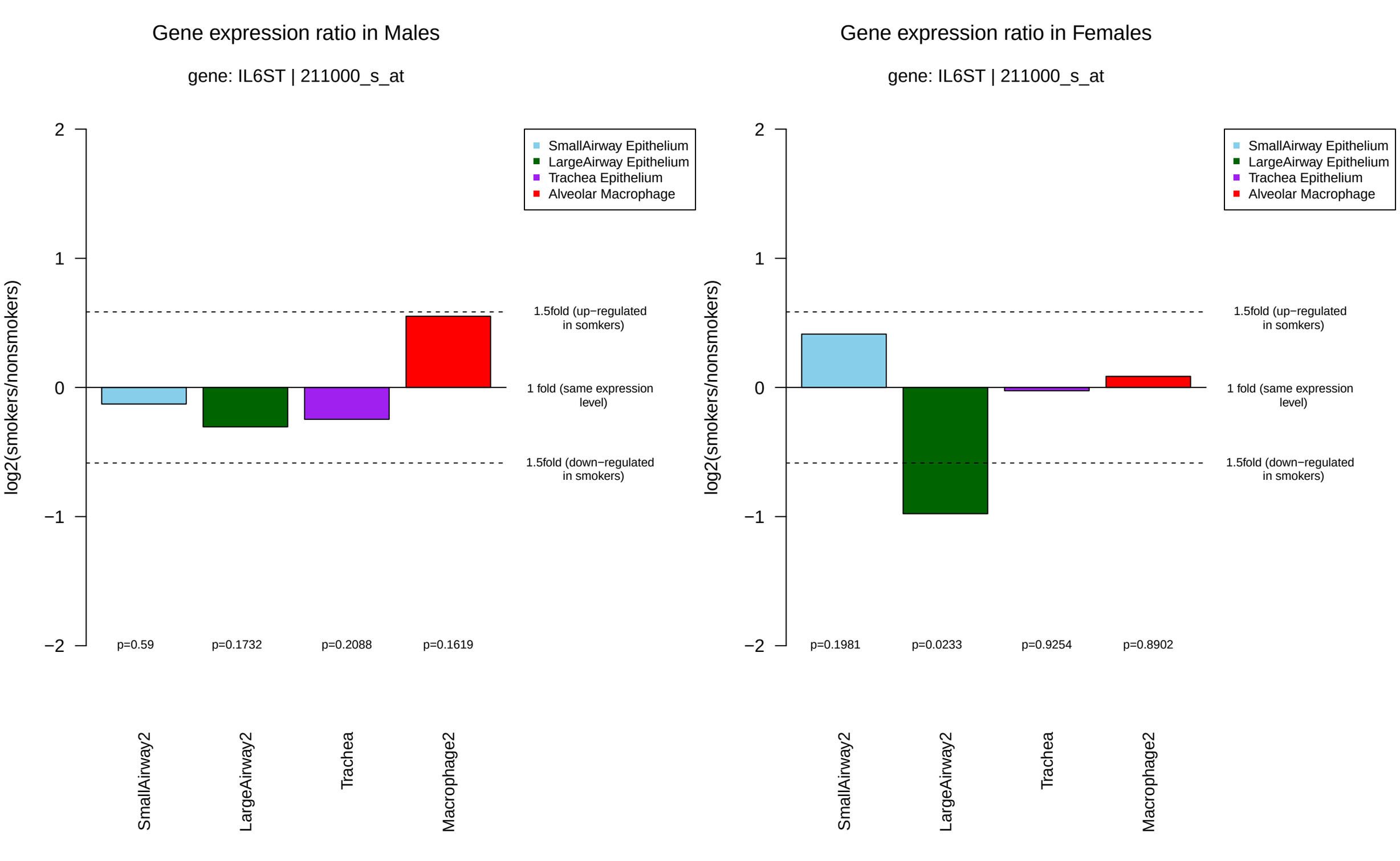
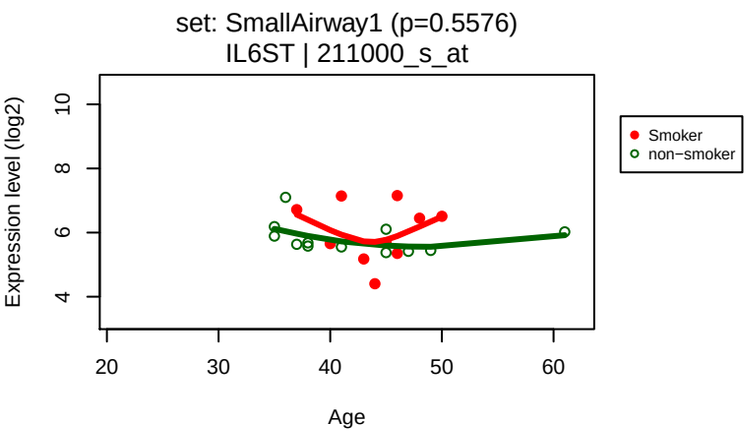
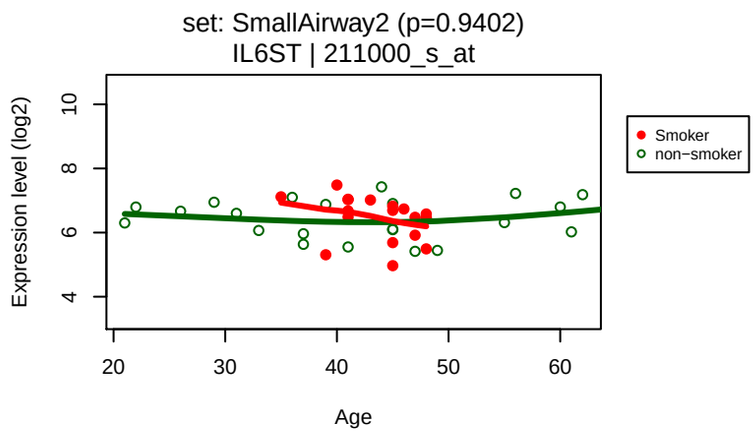
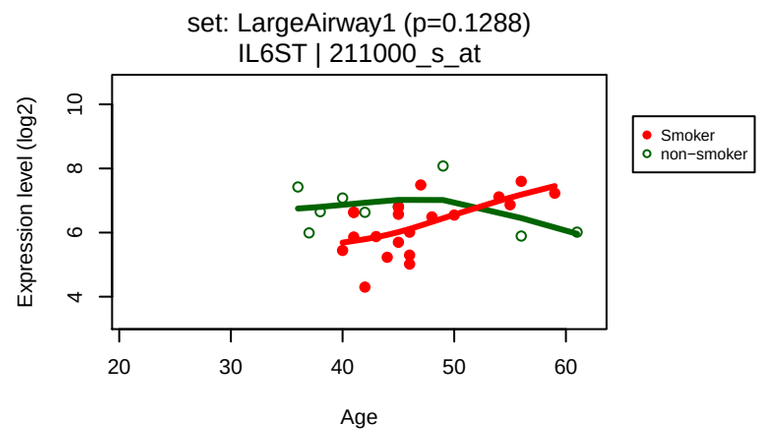

Gene expression of smokers and nonsmokers
gene: IL6ST | 211000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5576
p=0.9402
p=0.4728
p=0.1288
p=0.0053
p=0.1891
p=0.8648
p=0.2361
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

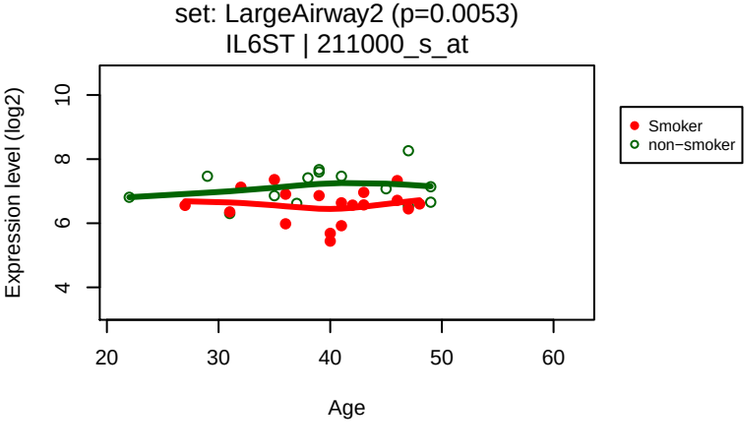

Gene expression ratio (smokers/nonsmokers)
gene: IL6ST | 211000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5576
p=0.9402
p=0.4728
p=0.1288
p=0.0053
p=0.1891
p=0.8648
p=0.2361
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

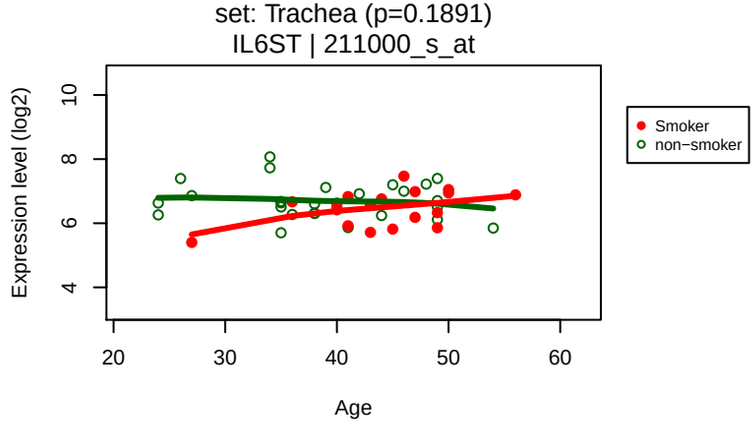

Gene expression ratio in Males
gene: IL6ST | 211000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.59
p=0.1732
p=0.2088
p=0.1619
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL6ST | 211000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1981
p=0.0233
p=0.9254
p=0.8902
SmallAirway2
LargeAirway2
Trachea
Macrophage2

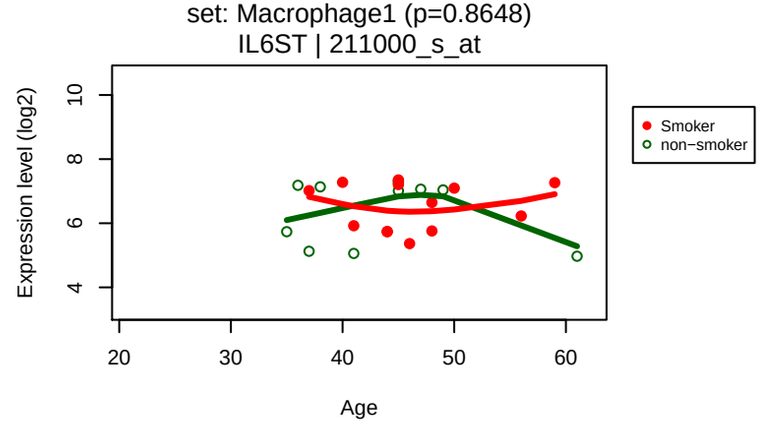

set: SmallAirway1 (p=0.5576)
IL6ST | 211000_s_at
Expression level (log2)
Age
Smoker
non−smoker

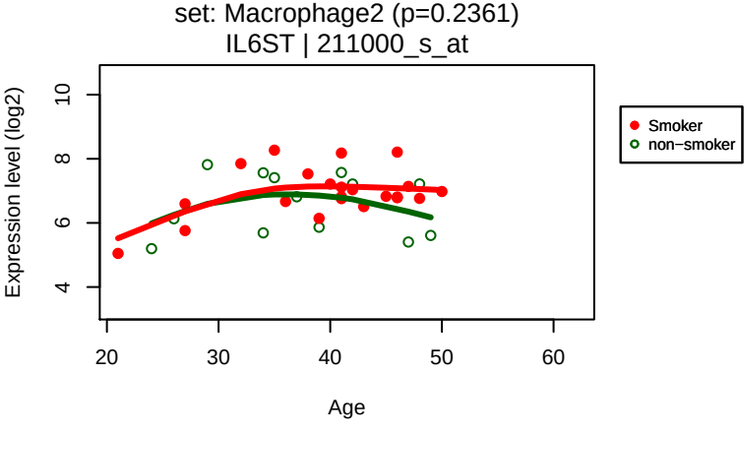

set: SmallAirway2 (p=0.9402)
IL6ST | 211000_s_at
Expression level (log2)
Age
Smoker
non−smoker

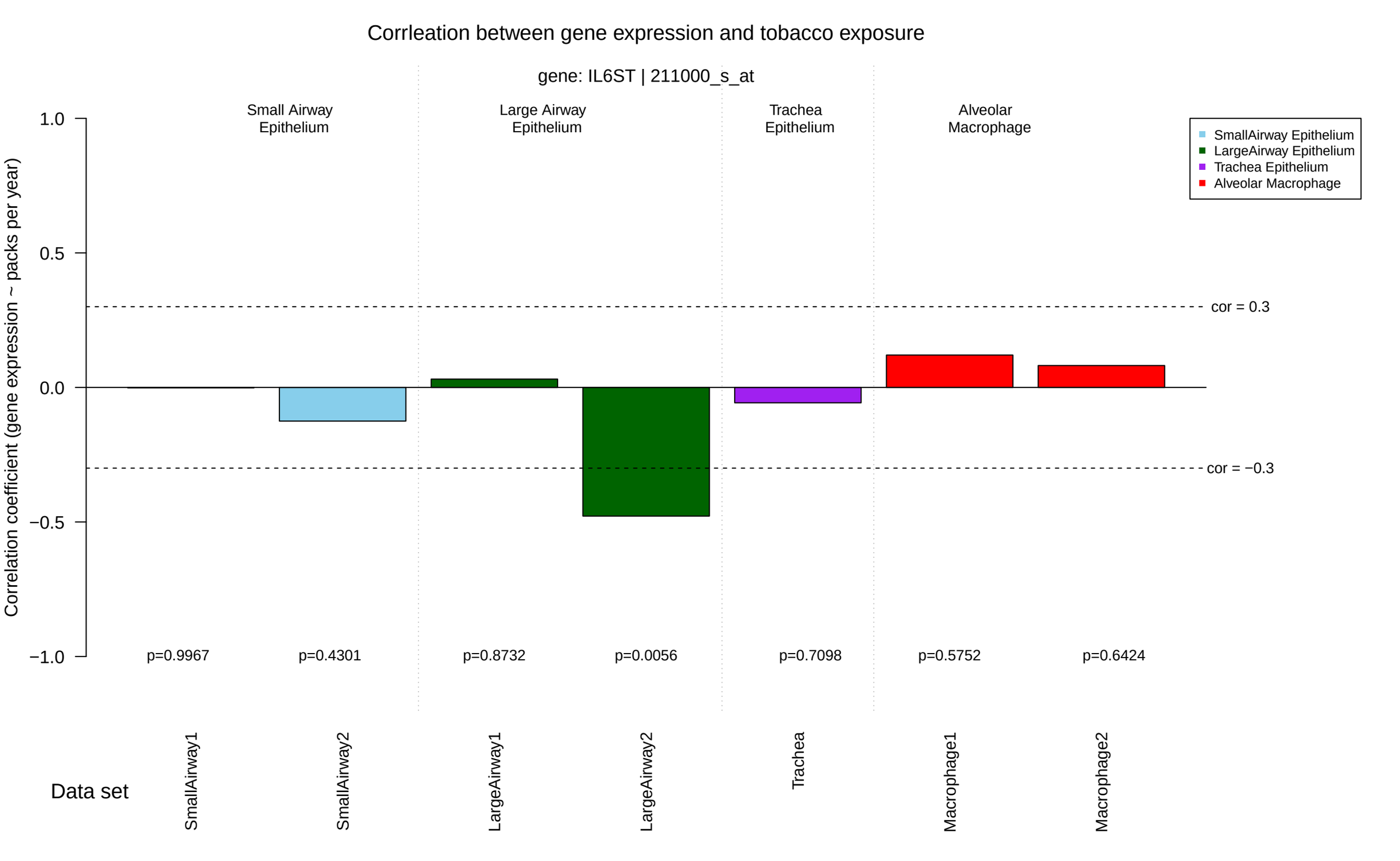

set: LargeAirway1 (p=0.1288)
IL6ST | 211000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0053)
IL6ST | 211000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1891)
IL6ST | 211000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8648)
IL6ST | 211000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2361)
IL6ST | 211000_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL6ST | 211000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9967
p=0.4301
p=0.8732
p=0.0056
p=0.7098
p=0.5752
p=0.6424
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2